MILITARY SPECIFICATION

FIRE SYSTEM, 20-MILLIMETRE POINT AND

SUPPRESSIVE HELICOPTER, AH-1T (TOW),

A/A49E-7(V4)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. This specification establishes the requirements for performance, design, test, manufacture, and acceptance of the A\A49E-7(V4) AH-1T (TOW) Helicopter 20-Millimetre (mm) Point and Suppressive Fire System, hereinafter referred to as the system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft" and Missiles

Beneficial comments (recaumendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self addressed Standardization Document improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-81O	Environmental Test Methods

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein.

DRAWINGS

General Electric Company l

105D3853	Adaptor, Recoil Mechanism
116C3555	Transducer, Pressure
152D5402	Interface, Slider Assembly
152D5403	Interface, Pressure Transducer
167C1557	Slider Assembly
167C2095	Coincidence Control Unit
189F788	Feed System, Ammunition
189F861	Torque Box Assembly
189F901	Gun Control Assembly

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DRAWINGS (continued)

197C7325	Interface, Coincidence Control
197C7326	Interface, Torque Box
201F525	Turret Control Assembly
201F798	Turret Assembly, 20-mm
201F799	Drive Assembly, Gun
209F115	20-mm Point and Suppressive Fire System
218F326	Interface, Gun Drive Assembly
218F327	Interface, Turret Control Assembly
218F328	Interface, Turret Assembly, 20-mm Gun
218F329	Interface, Gun Control Assembly
218F330	Interface, Recoil Adapter
218F331	Interface, Ammunition Feed System
798E556	Schematic Diagramt, Electrical System Interconnections, AH-lT (TOW)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specified acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Item description. The system shall be an electrically-powered turret that includes an ammunition storage and handling subsystem. The system shall also include the necessary controls and interfacing electronics for turret operation. The system shall accommodate an M197 20-mm three-barrel gun. The turret assembly and the gun shall provide a complete helicopter armament system. The major components to be manufactured shall be as specified in 3.1.3.

3.1.2 Interface definition.

3.1.2.1 Electrical interface

3.1.2.1.1 Electrical power supply. The system shall operate from power sources with characteristics conforming to MIL-STD-704, Category B. The electrical power for all testing shall be as specified in Table I.

TABLE I. Input electrical power.

Voltage	Current (Amperes)	Frequency (Hertz)
28 Vdc	125 (steady-state) 600 (peak)	--
115 ac	1	400

3.1.2.1.2 Schematic diagram. A schematic diagram of the electrical subsystem interconnection requirements is shown on Drawing 798E556.

3.1.2.2 Physical interface. The physical interface require=nts shall be in accordance with the drawings specified in Table 11.

TABLE 11. Physical interface requirements.

Component	Drawing
Turret Assembly	218F328
Gun Control Assembly	218F329
Gun Drive Assembly	218F326
Turret Control Assembly	218F327

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TABLE II. Physical interface requirements - Continued

Component	Drawing
Pressure Transducer	152D5403
Recoil Adapter	218F330
Ammunition Feed System	218F331
Slider Assembly	152D5402
Torque Box assembly	197C7326
Coincidence Control Box	197c7325

3.1.3 Major component list. The major components of the system, Drawing 209F115, shall be as specified in Table III.

TABLE III. Major component list.

Major Component	Drawing
Turret Assembly	201F798
Gun Control Assembly	189F901
Gun Drive Assembly	201F799
Turret Control Assembly	201F525
Pressure Transducer	116C3555
Recoil Adapter	105D3853

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TABLE III. Major component list - Continued

Major Component	Drawing
Ammunition Feed System	189F788
Slider Assembly	167C1557
Torque Box Assembly	189F861
Coincidence Control Box	167C2095

3.1.4 Government furnished property. The following equipment will be furnished by the Government for testing of the system (see 6.2.1):

a. M55A1B1 or M51A3 20-mm dummy ammunition

b. M14A1 ammunition links

c. M197 20-mm 3-barrel gun (barrels may be shortened to facilitate test)

d. M89E1 feeder

e. M55A2 target practice (TP) ammunition;

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Positioning accuracy. For any position within the turret angular coverage, the turret shall assume a position not greater than ± 5 roils in the elevation or azimuth axes from a commanded static position.

3.2.1.2 Dispersion. With the turret mounted on a test stand and zero error signal applied to the turret, the system shall fire a continuous 100-round burst that results in 80 percent of the rounds fired either falling within or cutting a circle with a maximum diameter of 12 mils.

3.2.1.3 Firing rates. With the turret in the positions specified in Table IV, the system shall fire at a rate of 675 + 30, -75 shots per minute (spin).

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TABLE IV. Fire test positions.

Position	Azimuth (Degrees)	Elevation (Degrees)
1	0 ± 2	0 ± 2
2	0 ± 2	-49 ± 1
3	+107 ± 2	+10 ± 1
4	-107 ± 2	0 ± 2
5	-107 ± 2	-49 ± 1

Note. Plus (+) values indicate a right azimuth or up elevation position. Minus (-) values indicate a left azimuth or down elevation position.

3.2.1.4 Clearing cycle. The gun shall be automatically cleared at the end of each firing burst. No live rounds shall be ejected during clearing. However, live rounds may be ejected during clearing if one of the following conditions occur:

a. The bore line of the firing barrel differs from the commanded bore line position in either azimuth or elevation by more than 5.5 degrees

b. There is a command to fire any fixed foward firing wing store

c. The turret is commanded to the normal stow position.

3.2.1.5 Firing interruption. The firing circuit shall be interrupted and the gun cleared in accordance with 3.2.1.4 when any of the following conditions occur:

a. After any actuation commnd to fire any fixed forward firing wing store, the actuation sequence shall intermpt firing and shall return the turret to the fixed forward firing position until the cmnmand is released. A time delay of 1.0 + 0.5 second shall be provided to assure safe projectile distance-before the turret can move out of the fixed forward firing position.

b. The bore line of the firing barrel differs from the commanded boreline position in either azimuth or elevation by more than 5.5 degrees.

c. The turret reaches the limit of its field of fire in elevation or azimuth.

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3.2.1.6 Turret angular coverage. Turret angular coverage shall be limited by both mechanical and electrical stops. The mechanical limit shall be defined as the first point of contact with the stop mechanism. The electrical limit shall be the point at which turret drive power is turned off and the gun is cleared.

3.2.1.6.1 Azimuth. The mechanical stops shall allow the turret to traverse 110 +1, -0 degrees in azimuth to the left and right of the straight forward position. The electrical stops shall allow the turret to traverse 109.50 + 0.25 degrees to the left and right of the straight forward position.

3.2.1.6.2 Elevation. The gun elevation limits shall be in accordance with Table V.

TABLE V. Turret elevation limits.

Azimuth (Degrees)	Elevation (Degrees)		
	Mechanical	Electrical 1/	Dynamic 2/
0 through 5	15.25	14.75	17.25
10	16.43	15.93	18.43
20	18.50	18.00	20.50
30	19.40	18.90	21.40
40	21.00	20.50	23.00
45.8	21.00	20.50	23.00
Straight Line to 110	12.21	11.71	14.21

1/ Electrical limit tolerance shall be +0.50 degrees and shall activate prior to contacting the mechanical limit.

2/ The dynamic limit shall be defined as the maximum turret saddle position that results from the following conditions. With the electrical limit disabled, the dynamic limit is reached when a fully equipped M197 gun *or* equivalent mass is installed in the turret and elevated at the full elevation slew velocity into the mechanical stop.

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3.2.1.6.3 Depression. The mechanical depression limit shall be -50 +1 degrees from a plane parallel to the turret mounting surface. The electrical depression limit shall be -50.25 + 0.25 degrees from a plane parallel to the turret mounting surface.

3.2.1.7 Turret velocity.

3.2.1.7.1 Angular velocity. Turret angular velocity (steady-state) shall not be less than 80 degrees per second in azimuth and not less than 60 degrees per second in elevation. The turret shall slew from the maximum left azimuth (-109 degrees) to the maximum right azimuth position (+109 degrees) in not more than 3.5 seconds and from the maximum right azimuth position (+109 degrees) to the maximum left azimuth position (-109 degrees) in not more than 3.5 seconds. The turret shall slew from the maximum elevation position to the depression angle of 50 degrees in not more than 1.5 seconds and from 50 degrees depression to the maximum elevation position in not more than 1.5 seconds.

3.2.1.7.2 Elevation velocity error. With the elevation command signal changing at a constant rate corresponding to a turret movement of five degrees per second in elevation, the turret shall follow this motion within three mils as measured by the synchro error signal received by the servo amplifiers.

3.2.1.7.3 Azimuth velocity error. With the azimuth command signal changing at a constant rate corresponding to a turret movement of five degrees per second in azimuth, the turret shall follow this motion within two mils as measured by the synchro error signal received by the servo amplifiers.

3.2.1.8 Turret acceleration. The turret drives shall accelerate the turret at a minimum rate of 120 degrees per second squared in any axis or combination of axes. The acceleration shall be measured using a step input from standstill.

3.2.1.9 Fixed turret positions.

3.2.1.9.1 Fixed forward firing. The turret shall assume the fixed forward firing position when a gunner or pilot command signal is received.

3.2.1.9.2 Upper stow. When the aircraft action switch is not depressed, the turret shall maintain an upper stow position with the gun at 0 ± 1 degree azimuth and elevated 11 + 1 degrees relative to the helicopter waterline. The turret shall assume and retain the upper stow position whenever the action signal is removed and there is no fixed forward firing command.

3.2.1.9.3 Emergency stow. If system electrical power is lost, the emergency power system shall return the turret to 0 + 5 degrees in azimuth and a minimum of 10 degrees in elevation relative to the aircraft datum line. When in the emergency stow position, the system shall provide a signal for a panel indication.

3.2.1.10 Helicopter wing stores safety interlock. The wing stores safety interlock shall prevent the firing of wing stores except when the turret gun line is within five degree's of the zero degrees azimuth position.

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3.2.1.11 Ammunition remaining signal. The ammunition remaining signal shall permit a count of the ammunition remaining in the storage and handling subsystem. The counting accuracy tolerance shall be ± 10 rounds until the last round has passed the counting device, after which the accuracy requirement shall not apply.

3.2.1.12 Ammunition totalizer. The ammunition totalizer shall indicate the total number of rounds that have been cycled through the system. The totalizer shall perform independently of the gun and shall count all rounds cycled including dummy and misfired rounds.

3.2.2 Physical characteristics.

3.2.2.1 Weight. The weight of the system shall be not greater than 350 pounds.

3.2.2.2 Ammunition storage capacity. The system, including the feed chute, shall have an ammunition storage capacity of 750 -10 rounds.

3.2.3 Reliability. The mature system shall have a reliability of 7500 mean rounds between failure (MRBF). See 6.3.3.1 and 6.3.3.2.

3.2.4 Maintainability.

3.2.4.1 Access. The design shall permit simple and rapid access to the major components without their removal from the system.

3.2.4.2 Gun replacement. The turret design shall allow the 1197 gun to be replaced by two persons in a maximum of 10 minutes. The required support equipment shall be available during gun replacement.

3.2.4.3 Turnaround time. A maximum of 15 minutes shall be required to fully reload and arm the system. When possible, the system design shall allow personnel under combat conditions to fully reload and arm the system without the use of any tool. Quick disconnect fasteners and hand-operated latches shall be used on access panels. Latches, pins, and any other type of fastener shall be operable by personnel wearing cold weather clothing. If the use of a screw-driver is required, the type of screw heads used shall be standardized.

3.2.4.4 Special tools. The system and its components shall be designed for assembly, disassembly, and installation using standard and commercial tools. The use of special tools shall be kept to a minimum.

3.2.5 Environmental conditions. The system shall be capable of meeting the performance requirements of this specification when subjected to the natural and induced environments specified in 3.2.5.1 through 3.2.5.7.

3.2.5.1 Altitude. The system shall be capable of meeting the performance requirements of this specification during exposure to a maximum altitude of 20,000 feet.

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3.2.5.2 Temperature. The system shall be capable of meeting the performance requirements of this specification during exposure to temperatures of -65°F to 125°F and after being subjected to temperatures of -80°F to 160°F for 72 hours.

3.2.5.2 Humidity. The system shall be capable of meeting the performance requirements of this specification after exposure to a maximum relative humidity of 100 percent including conditions where condensation takes place in the form of water or ice.

3.2.5.4 Salt fog. The system shall be capable of meeting the performance requirements of this specification after exposure to salt atmosphere of 5 ± 1 percent for 48 hours.

3.2.5.5 Dust. The system shall be capable of meeting the performance requirements of this specification after exposure to dust.

3.2.5.6 Vibration. The system shall be capable of meeting the performance requirements of this specification after being subjected to vibration on all three major axes in accordance with the conditions of Table VI,

TABLE VI. Vibration conditions.

Frequency (hertz)	Acceleration (± g)	Time per Axis (hours)	Total Time (hours)
10.6	0.75	2	6
21.2	1.00	1	3
31.8	0.75	1	3
42.4	1.00	1	3

Any new equipment added to the system as a result of redesign shall meet the requirements of MIL-STD-810, Method 514.2, Procedure I.

3.2.5.7 Shock. The system shall be capable of meeting the performance requirements of this specification after exposure to mechanical impulses of ± 4.5 g's in all axes.

3.2.6 Transportability. The system shall be efficiently transportable by normal transportation methods to permit employment, deployment and logistics support.

3.3 Design and constructions. The system shall have the design and construction characteristics specified herein.

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3.3.1 Materials, processes and parts. Particular attention shall be given to the selection of materials, processes, and parts to facilitate interchangeability stocking, and replacement in service. The number of different types and sizes of consumable bulk maintenance materials shall be kept to a minimum. Noncritical materials shall be used wherever practicable when performance, interchangeability, safety or reliability will not be adversely affected or production significantly altered. Specification and standards for materials, processes, and parts shall be selected in accordance with the order of precedence specified in MIL-STD-143.

3.3.2 Electromagnetic radiation. Except during gun firing, the system shall meet the requirements, procedures, and limits of MIL-I-6181. Any new electronic equipment added to the system as a result of redesign shall confomn to the requirements specified in the appendix.

3.3.3 Identification plates and product marking. Identification plates and product marking shall be in accordance with MIL-STD-130.

3.3.4 Interchangeability. Unless otherwise specified on the drawing, all parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable and replaceable with each other with respect to installation and performance in accordance with MIL-I-8500.

3.3.5 Safety. The system shall include fail-safe features for the safety of personnel during installation, operation, maintenance, and repair.

3.3.6 Human performance/human engineering. The principles and procedures of human engineering shall be applied to the system to assure efficient, safe, and reliable human performance.

3.3.7 Standards of manufacture. The system shall be manufactured in accordance with the standards specified on the applicable drawings.

3.3.8 Production drawings. The system shall be fabricated and assembled in accordance with the drawings, parts lists and other documents listed on Drawing 209F115.

3.4 Major component characteristics. The characteristics of the major components shall be in accordances with the drawings specified in Table III.

3.5 First article. When specified, a sample(e) shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.6 First article sample. When specified in the contract or purchase order, the contractor shall furnish a first article sample of one system for first article inspection and approval (see 6.2.1). The first article shall be manufactured using the same methods, materials, processes and procedures proposed for production.

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3.7 Workmanship. Finished items and parts shall not exhibit faulty material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, burrs, foreign matter, deformations, and missing operations that may affect serviceability, operation, or safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirments as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General. This section shall specify the methods used to confirm that the system as fabricated and assembled, complies with the requirements of this specification and the applicable assembly drawings.

4.1.1.1 Contractor's quality control system. The contractor shall provide and maintain an effective quality control system in accordance with the provisions of the contract (see 6.2.1).

4.1.1.2 Test provisions.

4.1.1.2.1 Verification category. Unless otherwise specified in the contract or purchase order (see 6.2.1), the requirements of Section 3 and 5 shall be verified in accordance with the test methods specified herein.

a. First article sample inspection (see 4.2.1)

b. Acceptance test (see 4.2.2)

4.1.1.2.2 Verification method. Each requirement of Sections 3 and 5 shall be verified in accordance with 4.3 by one or more of the following methods:

a. Inspection

b. Analysis

c. Demonstration

d. Test

4.1.1.2.2.1 Inspection. Verification by inspection shall consist of visual and dimensional checks. One or both may be necessary to verify a requirement.

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4.1.1.2.2.2 Analysis. Analysis shall consist of applying empirical/ theoretical relationships between conditions, facts and test data to predict compliance with the specified requirement. The use of previous test data for verification by similarity of design shall be permitted where applicable.

4.1.1.2.2.3 Demonstration. Demonstration shall be a fit and function check, usually of a go/no-go nature that requires no instrumentation.

4.1.1.2.2.4 Test. Test shall consist of making observations or measurements on a quantitative basis for compliance with the specified requirement and normally requires instrumentation. If an instrumentation/equipment failure occurs, the test will be declared "no test" and the test repeated.

4.1.1.2.3 Test conditions. Unless otherwise specified herein, all tests shall be performed under the following conditions:

a. Temperature: 70°F ± 20"F, except firing tests may be performed at ambient conditions

b. Barometric pressure: 28 to 32 inches of mercury

c. Humidity: Up to 95 percent

4.1.2 Special tests and examinations. Not applicable.

4.2 Quality conformance inspections.

4.2.1 First article inspection.

4.2.1.1 Full first article inspection. When specified in the contract or purchase order (see 6.2.1), the first article sample of 3.6 shall be subjected to ail the tests and examinations specified in 4.3.

4.2.1.2 Abbreviated first article inspection. When specified in the contract or purchase order (see 6.2.1) and if the contractor has previously manufactured the system, only the first article inspections of Table VII shall be performed.

TABLE VII. Abbreviated first article inspections.

Requirement	Title	Test Procedure
	Acceptance test	4.2.2
3.1.2.2	Physical interface	4.3.1.2.2
3.2.1.3	Firing rate	4.3.2.l.3a

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4.2.1.3 First article acceptance and rejection criteria. Failure of the system to pass any of the first article inspections specified herein shall be cause for rejection of the first article sample.

4.2.1.4 First article test plan. A contractor proposed test plan shall be made available for review and acceptance by the procuring activity (see 6.2.1).

4.2.2 Acceptance test. The acceptance test to be performed on each system shall be limited to those requirements and procedures specified in Table VIII.

TABLE VIII. Acceptance test.

Requirement	Title	Test Procedure
3.1.2.1.1	Electrical power supply	4.3.1.2.1.1
3.1.3	Major component list	4.3.1.3
3.2.1.1	Positioning accuracy	4.3.2.1.1
3.2.1.2	Dispersion	4.3.2.1.2
3.2.1.3	Firing rates	4.3.2.1.3
3.2.1.4	Clearing cycle	4.3.2.1.4
3.2.1.5	Firing interruption	4.3.2.1.5
3.2.1.6	Turret angular coverage	4.3.2.1.6
3.2.1.6.1	Azimuth	4.3.2.1.6
3.2.1.6.2	Elevation	4.3.2.1.6
3.2.1.6.3	Depression	4.3.2.1.6
3.2.1.7.1	Angular velocity	4.3.2.1.7
3.2.1.7.2	Elevation velocity error	4.3.2.1.7
3.2.1.7.3	Azimuth velocity error	4.3.2.1.7
3.2.1.8	Turret acceleration	4.3.2.1.8

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TABLE VIII. Acceptance test - Continued

Requirement	Title	Test Procedure
3.2.1.9.1	Fixed forward firing	4.3.2.1.9
3.2.1.9.2	Upper stow	4.3.2.1.9
3.2.1.9.3	Emergency stow	4.3.2.1.9
3.2.1.10	Helicopter wing stores safety interlock	4.3.2.1.10
3.2.1.11	Ammunition remaining signal	4.3.2.1.11
3.2.1.12	Ammunition totalizer	4.3.2.1.12
3.2.2.1	Weight	4.3.2.2.1
3.2.2.2	Ammunition storage capacity	4.3.2.2.2
3.3.3		4.3.3.3
3.3.4		4.3.3.4
3.3.7	Standards of manufacture	4.3.3.5
3.3.8	Production drawings	4.3.3.6
3.7	Workmanship	4.3.6
5.1	Preservation, packaging, packing, and marking	4.3.5

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4.2.2.1 Final examination. Each system shall have a final examination after completion of all testing and prior to preservation and packaging. Each system shall be visually examined for cleanliness, completeness of manufacture and workmanship.

4.2.2.2 Test nonconformance. If the test requirements cited herein are not met, acceptance shall be deferred until a corrective action plan is developed, implemented and required successful retests are accomplished. Failures shall be as defined in 6.3.3.1 and 6.3.3.2. The contractor shall accomplish the following:

a. Conduct a failure analysis performing dimensional, physical and visual examination of the components suspected to be the cause of the failure or malfunction.

b. Examine complete assemblies, partial assemblies and components (in stock, in-process, or at final assembly) to ensure that material containing the same potential for failure or malfunction is purged from the inventory and not presented for acceptance.

c. Correct the deficiency by removal of the identified cause of the failure or malfunction and restore with acceptable material and determine the retesting required.

d. Evaluate and correct the applicable production processes and procedures to preclude recurrence of the same failure or malfunction.

e. Coincident with implementing the corrective action, make available the findings of the failure analysis to the procuring activity.

f. Submit the unit to an acceptable retest cycle to verify the corrective action.

4.3 Test methods and procedures.

4.3.1 Item definition. The requirements of 3.1 shall be verified by successful completion of the inspections, demonstrations, and tests specified in 4.3.1.2 through 4.3.6.

4.3.1.2 Interface definition.

4.3.1.2.1 Electrical interface.

4.3.1.2.1.1 Electrical power supply. The electrical power inputs shall be measured and recorded during testing. The electrical power inputs shall be in accordance with 3.1.2.1.1.

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4.3.1.2.1.2 Schematic diagram. The requirements of 3.1.2.1.2 shall be verified by sucessful demonstration of the quality conformance inspection defined in 4.2.

4.3.1.2.2 Physical interface. The requirements of 3.1.2.2 shall be verified by inspection.

4.3.1.3 Major component list. The requirements of 3.1.3 shall be verified by inspection of the major components that are used during final assembly of each system.

4.3.1.4 Government furnished property. The contracting officer should arrange to furnish the property listed in 3.1.4.

4.3.2 Characteristics.

4.3.2.1 Performance.

4.3.2.1.1 Positioning accuracy. The requirements of 3.2.1.1 shall be verified by test. The test shall be in accordance with the following requirements:

a. A baseline position of zero degrees in elevation and azimuth shall be established.

b. A minimum of five static position shall be tested as follows:

(1) Zero degrees azimuth, zero degrees elevation

(2) Zero degrees azimuth, +11 degrees elevation

(3) Zero degrees azimuth, -30 degrees elevation or -50 degrees elevation

(4) Forty-five degrees right azimuth, zero degrees elevation

(5) Forty-five degrees left azimuth, zero degrees elevation

4.3.2.1.2 Dispersion. The requirements of 3.2.1.2 shall be verified by test of one system per month or one system in 10, whichever is greater. With the turret mounted on a test stand and zero error signal applied, a minimum burst of 100 TP rounds shall be fired at a target located 1000 inches from the muzzle.

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4.3.2.1.3 Firing rates. The requirements of 3.2.1.3 shall be verified during first article and quality conformance inspections as follows:

a. First article - A full complement of dummy ammunition shall be cycled through the first article sample to verify the minimum firing rate and test positions of 3.2.1.3.

b. Quality conformance - One system per month or one system in 10, whichever is greater shall be fire tested with the turret in the zero azimuth and zero elevation position to verify the firing rate of 3.2.1.3. A full complement of dummy ammunition shall be cycled through each system to verify the minimum firing rate of 3.2.1.3.

4.3.2.1.4 Clearing cycle. The requirements of 3.2.1.4 shall be verified during the test of 4.3.2.1.3. Visual observation and instrumentation shall be utilized to determine the clearing cycle function.

4.3.2.1.5 Firing interruption. The requirements of 3.2.1.5 shall be verified by test.

4.3.2.1.6 Turret angular coverage. The requirements of 3.2.1.6 shall be verified by demonstration. The turret shall be operated to the mechanical and electrical limits of azimuth, elevation, and depression.

4.3.2.1.7 Turret velocity. The requirements of 3.2.1.7 shall be verified by test. The turret angular velocity and the elevation and azimuth velocity errors shall be verified.

4.3.2.1.8 Turret acceleration. The requirements of 3.2.1.8 shall be verified by test.

4.3.2.1.9 Fixed turret positions. The requirements of 3.2.1.9 shall be verified by test. The fixed forward firing position, upper stow position, and emergency stow position shall be verified.

4.3.2.1.10 Helicopter wing stores safety interlock. The requirements of 3.2.1.10 shall be verified by test.

4.3.2.1.11 Ammunition remaining signal. The requirements of 3.2.1.11 shall be verified during the cycling test of 4.3.2.1.3.

4.3.2.1.12 Ammunition totalizer. The requirements of 3.2.1.12 shall be verified during the cycling test of 4.3.2.1.3.

4.3.2.2 Physical characteristics.

4.3.2.2.1 Weight. The requirements of. 3.2.2.1 shall be verified by inspection.

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4.3.2.2.2 Ammunition storage capacity. The requirements of 3.2.2.2 shall be verified by demonstration during the firing rate tests of 4.3.2.1.3.

4.3.2.3 Reliability. The requirement of 3.2.3 shall be verified by analysis. The analysis shall be made available for review and acceptance by the procuring activity (see 6.2.1).

4.3.2.4 Maintainability. The requirements of 3.2.4 shall be verified by analysis. The analysis shall be made available for review and acceptance by the procuring activity (see 6.2.1).

4.3.2.5 Environmental conditions. One complete system shall be used to support the environmental tests specified herein. Unless otherwise specified herein, the tests shall be performed at the system level. The system shall be placed in the applicable test chamber with an orientation similar to its installation in the helicopter. All cover fairings and inspection plates shall be in place.

4.3.2.5.1 Altitude-temperature. The requirements of 3.2.5.1 and 3.2.5.2 shall be verified by test in accordance with MIL-STD-810, Method 504.1, Procedure I, for Equipment Category 4. Intermittent operation shall not be required.

4.3.2.5.2 Humidity. The requirements of 3.2.5.3 shall be verified by test in accordance with MIL-STD-810, Method 507.1, Procedure I. The temperature cycle shall be from 100°F to 122°F.

4.3.2.5.3 Salt fog. The requirements of 3.2.5.4 shall be verified by test in accordance with MIL-STD-810, Method 509.1.

4.3.2.5.4 Dust. The requirements of 3.2.5.5 shall be verified by test in accordance with MIL-STD-810, Method 510.1, Procedure I. The test shall be performed at 77°F with wind-blown dust at a velocity of 300 ± 250 feet per minute.

4.3.2.5.5 Vibration. The requirements of 3.2.5.6 shall be verified by test. All three major axes shall be exposed to the vibration requirements of 3.2.5.6. The ammunition feed subsystem shall be loaded with 560 rounds for exposure to the vibration environment.

4.3.2 .5.6 Shock. The requirements of 3.2.5.7 shall be verified by test in accordance with MIL-STD-810, Method 516.2, Procedure I.

4.3.2.6 Transportability. The requirements of 3.2.6 shall be verified by analysis. The analysis shall be made available for review and acceptance by the procuring activity (see 6.2.1).

4.3.3 Design and construction.

4.3.3.1 Materials, processes, and parts. The requirements of 3.3.1 shall be verified by analysis and inspection of materials and components.

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4.3.3.2 Electromagnetic interference. The requirements of 3.3.2 shall be verified by test.

4.3.3.3 Identification plates and product marking. The requirements of 3.3.3 shall be verified by inspection of component and part marking.

4.3.3.4 Interchangeability. The requirements of 3.3.5 shall be considered verified by random selection of components and parts with no selective assembly allowed during normal production assembly.

4.3.3.5 Safety. The requirements of 3.3.6 shall be verified by analysis, demonstration, and inspection. The analysis shall be made available for review and acceptance by the procuring activity (see 6.2.1).

4.3.3.6 Human performance/human engineering. The requirements of 3.3.7 shall be verified by analysis. The analysis shall be made available for review and acceptance by the procuring activity (see 6.2.1).

4.3.3.7 Standards of manufacture and production drawings. Unless otherwise specified herein, the contractor shall make available to the procuring activity, records that each inspection assures conformance to the production drawing and standards of manufacture.

4.3.4 Major component characteristics. The requirements of 3.4 shall be verified by successful completion of the inspections and tests specified in this specification.

4.3.5 Preparation for delivery. Packaging and marking testing or inspection shall be as specified in the contract (see 6.2.1).

4.3.6 Workmanship. The requirements of 3.7 shall be verified by visual inspection of all system parts and assemblies.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Systems shall be preserved, packaged, packed and marked as specified in the contract (see 6.2.1).

6. NOTES

6.1 Intended use. The system is intended for use on the AH-1T helicopter.

6.2 Ordering data.

6.2.1 Acquisition documents. Acquisition documents shall specify the following:

a. Title, number, and date of this specification

b. Government furnished property (see 3.1.4)

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c. Preservation, packaging, packing and marking requirements

d. First article sample (see 3.6)

e. Verification method if other than specified in 4.1.1.2.2

f. First article test plan (see 4.2.1.4)

g. Reliability analysis (see 4.3.2.3)

h. Maintainability analysis (see 4.3.2.4)

i. Transportability analysis (see 4.3.2.6)

j. Safety analysis (see 4.3.3.5)

k. Human performance analysis (see 4.3.3.6)

l. Delivery instructions

m Additions to this specification

n. Exceptions to this specification.

6.2.2 Data requirements. When this specification is used in an acquisition that incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.	Option
4.2.1.4	First Article Test Plan	DI-T-4901	____
4.3.2.3	Reliability	DI-R-7079	____
4.3.2.4	Maintainability	DI-R-7109	____
4.3.2.6	Transportability	DI-L-3327A	____
4.3.3.5	Safety	DT-H-7049	____
4.3.3.6	Human performance/human engineering	DI-H-7054	____

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(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Contractor. The organization that contracts to manufacture the system.

6.3.2 Procuring activity. The organization that contracts to buy the system.

6.3.3 Failure.

6.3.3.1 Chargeable system failure. A chargeable system failure is defined as follows:

a. Failure to operate on signal to fire

b. Failure to cease operation on signal to stop firing

c. Failure to continue operation while the fire signal is applied

d. Uncommanded start

e. Failures that prevent the system from completing its mission.

6.3.3.2 Nonchargeable failure. Failures attributable to any of the following should not be considered a chargeable failure:

a. Failure of test equipment or government furnished equipment (GFE)

b. Failure caused by personnel errors

c. Failure of associated equipment

d. Dependent (secondary) failure

e. Failures eliminated by corrective action

f. Ammunition or link failures that prevent firing or cause the system to stop.

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6.4 First article. When a first article inspection is required, the item will be tested and should be the first production item. The first article should consist of one unit. The contracting officer should include specific istructions in acquisition documents regarding arrangements for examination and test approval of the first article.

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APPENDIX

ELECTROMAGNETIC RADIATION REQUIREMENTS

10. SCOPE

10.1 This appendix establishes the electromagnetic radiation requirements for any new electronic equipment that is added to the system as the result of a redesign. This appendix is a mandatory part of the specification for any new electronic equipment that is added to the system as the result of a redesign. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks" Unless Otherwise specified, the following specifications, Standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this appendix to the extent specified herein.

SPECIFICATIONS

Military

.MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems
MIL-E-6051	Electromagnetic Compatibility Requirements, systems

STANDARDS

Military

MIL-STD-454	Standard General Requirements for Elctronic Equipment
MIL-STD-461	Electranagnetic Emission and Susceptibility Requirements for the Control of Electraagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-1385	Preclusion of Ordnance Hazards in Electromagnetic Fields, General Requirements for

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APPENDIX

a. Tests CE01 (all lines), RE01, and RSO1 shall be performed for information only.

b. The upper limits of CS03, CS04, CS05 and RE02 shall be the maximum RF plus the maximum IF. The separation of test signals in CS03, CS04, and CS05 shall be tailored to effectively test the receiver.

c. The CS02 requirement shall be applicable to all neutrals and grounds.

d. Interconnecting signal leads shall be tested to the same limits as interconnecting control leads in test CE03.

e. The maximum duration of the conducted switching transients in test CE07 is 50 µs. Transients longer than 50 us shall be within the limits of Figures 5 and 10 of MIL-STD-704. Test CE07 shall be performed without the external 10 UF line filters.

f. Thexxnaximum signal generator levels in test CS03 are +10 dBm.

g. The placement of the cable used to generate a magnetic field for RS02 shall be as shown in Figure 7a of MIL-STD-462.

h. The RS03 levels are those of the MIL-HDBK-235 environment reduced by estimated airframe shielding.

i. The test limits below 2 MHz are [50 + 42 log (2000/F)] for CE03 broadband emissions. F is in KHz.

j. The spike limits for the CS06 and RS02 requirements are:

E2 = 180 V, t2 less than or equal to 0.15 us

k. The RE02 limits shall extend linearly for the higher frequencies.

l. me CS02 test source shall be 3V instead of 1V.

m. The RS04 test of MIL-STD-462 shall be performed for frequencies below 2 MHz.

n. The CE03 from 15 KHz to 20 KHz can be relaxed for lines with load currents greater than one ampere. The value at 15 KHz can be relaxed by 20 log (load current). The new limits are then a line drawn from the new value at 15 KHz to the present value at 20 KHz.

o. The RS03 test limits shall be as specified in Table IX.

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APPENDIX

a. Tests CE01 (all lines), RE01, and RS01 shall be performed for information only.

b. The upper limits of CS03, CS04, CS05 and RE02 shall be the maximum RF plus the maximum IF. The separation of test signals in CS03, CS04, and CS05 shall be tailored to effectively test the receiver.

c. The CS02 requirement shall be applicable to all neutrals and grounds.

d. Interconnecting signal leads shall be tested to the same limits as interconnecting control leads in test CE03.

e. The maximum duration of the conducted switching transients in test CE07 is 50 μs. Transients longer than 50 us shall be within the limits of Figures 5 and 10 of MIL-STD-704. Test CE07 shall be performed without the external 10 μF line filters.

f. The maximum signal generator levels in test CS03 are +10 dBm.

g. The placement of the cable used to generate a magnetic field for RS02 shall be as shown in Figure 7a of MIL-STD-462.

h. The RS03 levels are those of the MIL-HDBK-235 environment reduced by estimated airframe shielding.

i. The test limits below 2 MHz are [50 + 42 log (2000/F)] for CE03 broadband emissions. F is in KHz.

j. The spike limits for the CS06 and RS02 requirements are:

E2 = 180 V, t2 less than or equal to 0.15 μs

k. The RE02 limits shall extend linearly for the higher frequencies.

l. The CS02 test source shall be 3V instead of 1V.

m. The RS04 test of MIL-STD-462 shall be performed for frequencies below 2 MHz.

n. The CE03 from 15 KHz to 20 KHz can be relaxed for lines with load currents greater than one ampere. The value at 15 KHz can be relaxed by 20 log (load current). The new limits are then a line drawn from the new value at 15 KHz to the present value at 20 KHz.

o. The RS03 test limits shall be as specified in Table IX.

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APPENDIX

Table IX. RS03 test limits.

Freq. Range (MHz)	Equipment Bays Field Intensity (volts/meter)		Other Equipment Areas Field Intensity (volts/meter)	
	Required	Goal	Required	Goal
0.014-200	20	100	200	---
200-450	65	100	200	650
450-1000	20	100	200	---
1000-12,000	100	---	200	---
12,000-40,000	2	---	2	---
16,500	100	---	200	2500
33,000	100	---	200	4000

30.3 Wiring and cabling. all interconnecting cables shall be classified into groupings representing categories of similar signals. The contractor shall provide each type with shielding and grounding criteria that are in accordance with sound E^3 design procedures. Maximum spacings between dissimilar cable harnesses shall be maintained and harnesses shall be routed as closely to the ground plane as permitted.

30.4 Degrdation criteria. Degradation criteria shall be established for the system. This criteria shall be used to predict and to evaluate malfunction and unacceptable responses. These criteria shall be defined in the system E^3 test procedures in terms of input and output parameters.

30.5 Safety margins. Safety margins shall be established for the system assigned to critically categories I and II. The safety margin for aircraft electro-explosive devices (EEDs) is defined by MIL-STD-1385.

30.6 Electrical power. The contractor shall provide E^3 design criteria to ensure the system installation shall not malfunction nor have unacceptable responses when supplied with electrical power conforming to MIL-STD-704. This specifically includes surges, ripples and other electrical conditions that can cause interference Short duration, infrequent transients or equipment wires are required to meet MIL-E-6051 spike requirements

30.7 Lightning protection. The contractor shall prvide E^3 design criteria to ensure that the system shall be designed to provide lightning protection in accordance with MIL-E-8057. The contractor shall make available for review and acceptance by the procuring activity a plan to demonstrate that all elements of the system shall be adequately protected against lightning hazards. Protection of personnel, fuel, weapons and ordnance subsystems shall be given special emphasis and proven by test. Lightning protection shall be

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provided to (a) prevent temporary upset and permanent damage to the avionics suite, (b) prevent structural damage to the airframe and (c) prevent lightning ignition of fuel tanks and systems.

30.8 Precipitation static. The contractor shall provide E design criteria to ensure that the system shall be designed to prevent P-static charges from degrading system effectiveness. Precipitation static control measures shall be included only where analysis or tests show that such measures are required and resistivity shall be that necessary to eliminate unacceptable responses. Static discharges shall be used only if analysis or tests show they are required.

30.9 HERO. The system design shall include provisions to protect ordnance subsystems from inadvertent ignition or dudding caused by any form of electromagnetic or electrostatic energy. All wiring, cabling and hardware associated with the ordnance subsystems shall be carefully designed to prevent stray pickup and eliminate undesired energy. All modes of operation during the mission, including loading, unloading, checkout, and prelaunch shall be considered.

30.10 RADHAZ. The system design criteria shall include provisions for protection of personnel from RF, electromagnetic, electrostatic and shock hazards in accordance with Requirement 1 of MIL-STD-454. Where possible, protection provisions shall be designed into associated subsystems/equipments. When protection by design is not technically feasible, adequate safety precautions shall be included in operating and maintenance manuals.

30.11 Operational EKE. The system design shall provide compatibility with the external electromagnetic environment (EKE). Consideration shall be given to the intended mission profiles, the available electromagnetic environment data and the degree to which the external environment can reduce the desired effectiveness of the system. The aircraft and its subsystems shall meet all design requirements when immersed in a friendly (carrier deck) or hostile external electromagnetic environment. Mission and safety requirements shall be met during operations when electromagnetic energy originating from friend or foe is coupled to the aircraft through CFE, GFE, GSE and external electrical power. Electromagnetic-environmental data is available from MIL-HDBK-235 and MIL-HDBK-253.

30.12 Emission control. The system design shall include an emission control function which prevents electromagnetic radiation from the system in excess of -110 dBm/M^2 at 1 nautical mile.

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